Car Security and Monitoring System

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***Abstract*—This project addresses the safety and security of vehicles. The project's main goal is to emergency response if the car crashes and remote monitoring of the car. We implemented these features fuel lead detection, car crash detection, and remote monitoring of the car. These features will prevent oil theft and secure the car system. As we already know, oil theft is very common. Our project not only prevents oil theft of the car but also provides safety for the car.**

***Index Terms*—Car Monitoring, Car security, Oil theft, Collision detection, IoT.**

I. Introduction

This project is designed to develop a complete Smart Car Security and Monitoring System that combines modern sensors and communication systems, in response to the growing concern of fuel robbery and street security [1]. Major objectives include live detection of unauthorized fuel access, crash detection, with an automated SOS call, and remote monitoring of the vehicle's speed and status. Overcoming these challenges will give vehicle owners improved security, less fuel loss, and peace of mind, while helping to ensure safe driving.

Additionally, our solution includes comprehensive monitoring features, allowing owners to access real-time data from their vehicles remotely, including speed, location, and vehicle status. Fuel theft risks have been reduced by the swiftly notifying of the owner when the fuel lid is opened by an unauthorized person. In case of collision, the sudden deceleration is detected through a gyroscope sensor [2]. On the LCD display on board the vehicle, sensor information, vehicle speed and diagnostic alerts are displayed. They are also available wirelessly through a mobile interface.

II. Proposed Method

The proposed architecture integrates sensory, control, and telematics layers to achieve automated monitoring and vehicle protection [3]. The central processing unit is a NodeMCU Mega WiFi R3 (combining an ATmega2560 and an ESP8266 module with 32 Mb flash memory). The system communicates wirelessly with the Blynk IoT application via Wi-Fi. Directional motor controls and alert widgets are mapped to virtual application pins [4]. Actuation is managed through an L298N motor driver powering four DC gear motors, powered by a dual-cell 7.2 V battery supply.

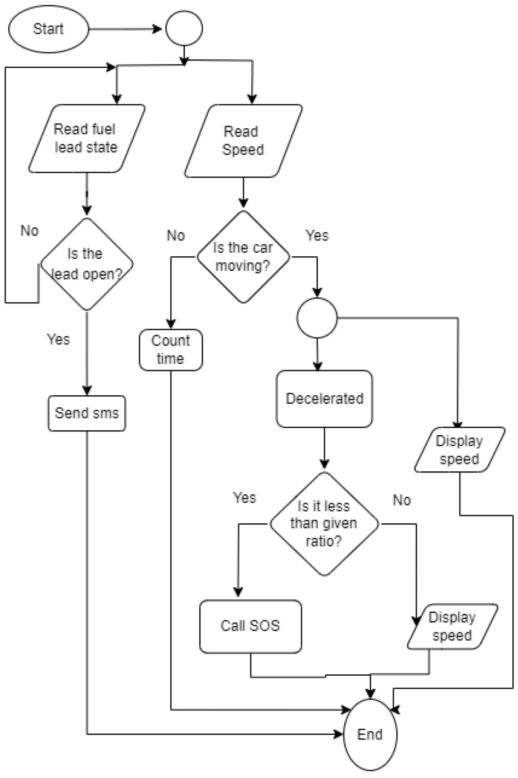


Fig. 1. Full system flowchart of the proposed Car Security and Monitoring System.

The overall flow of the proposed Car Security and Monitoring System is shown in Figure 1. The system is continually tracking the fuel lid status, vehicle speed, and abrupt changes in vehicle motion. An alert is displayed when the fuel lid is opened to alert the vehicle owner. The system also monitors the movement and braking of the vehicle(s) to detect possible collisions and generate emergency responses. This integrated monitoring process seeks to enhance vehicle safety and security.

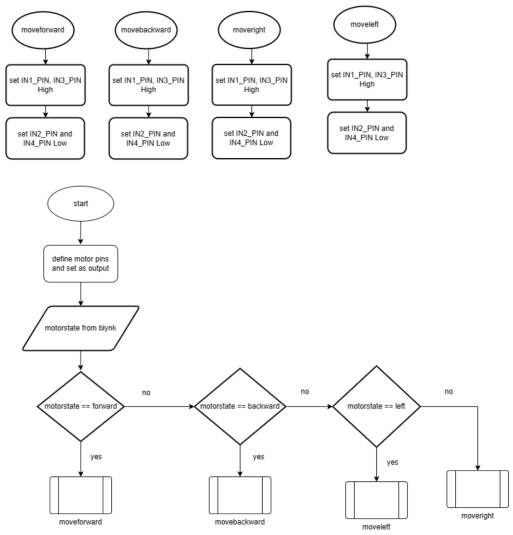


Fig. 2. Motor control subsystem diagram.

Figure 2 shows the motor control system for controlling vehicle movement. Directional commands are sent to the system via the Blynk app, and controlled by the NodeMCU Mega controller. The motor driver controls the DC gear motors based on instructions received to move the vehicle forward, backward, left or right. This is a wireless vehicle operating system that works by using a wi-fi network.

A Hall Effect sensor is used to measure the speed of the vehicle and is shown in Figure 3. The sensor creates pulses when the magnetic field changes due to the rotation of the wheel [5]. These pulses are recorded during a certain period of time and used to determine the rate of rotation of the wheel. The calculated value is then used to calculate the vehicle's speed which is displayed on the LCD screen for real time monitoring.

The collision detection process with the use of the MPU-6050 sensor is shown in the Figure 4. The sensor keeps tracking the vehicle acceleration changes on the X, Y and Z axis. The system detects an imminent collision when the measured acceleration is above a specified limit and alerts the user. The feature helps to detect and prevent accidents, enhancing vehicle safety.

The fuel lid monitoring mechanism using micro limit switch [1], as shown in Figure 5 , was designed. Switch detects the situation of the gas cap of the vehicle. When the lid is opened the system updates its status and notifies you with a message on the Blynk app. The system shows the appropriate status

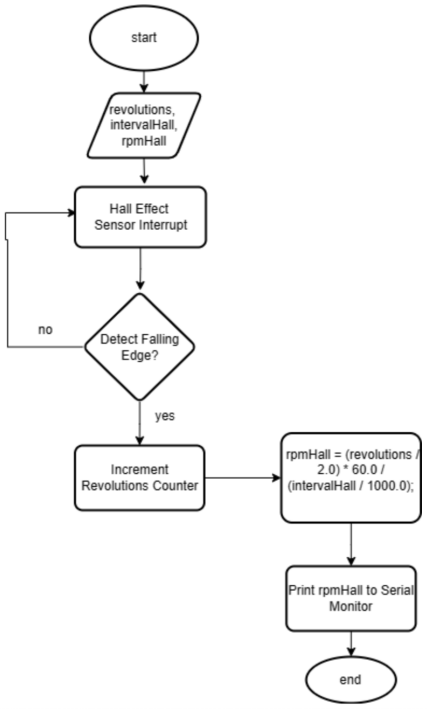


Fig. 3. Vehicle speed measurement flowchart using Hall Effect sensor.

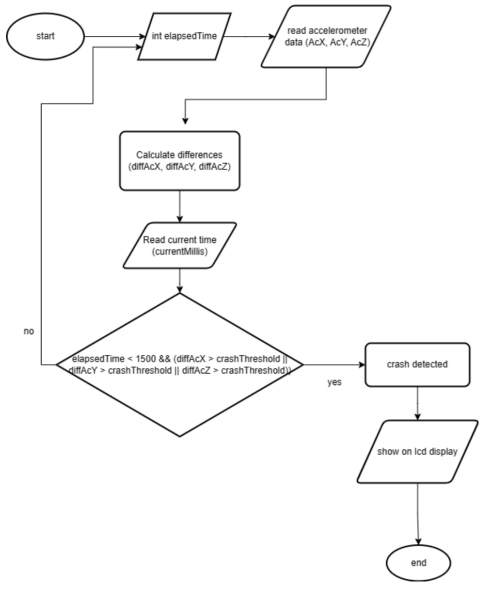


Fig. 4. Collision detection flowchart using MPU-6050 sensor.

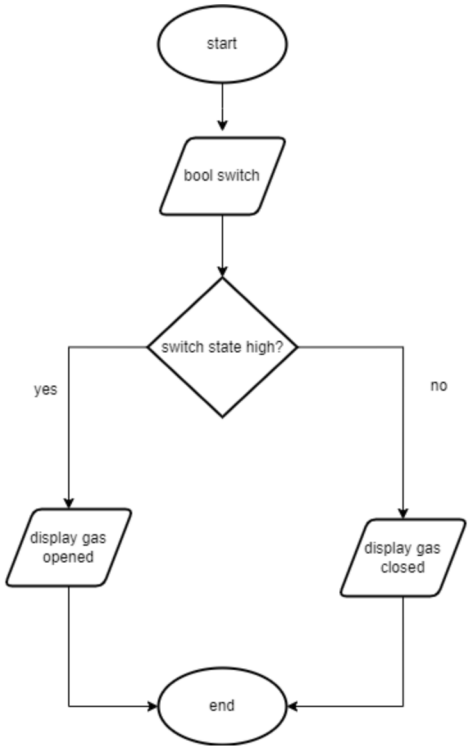


Fig. 5. Fuel lid monitoring flowchart using micro limit switch.

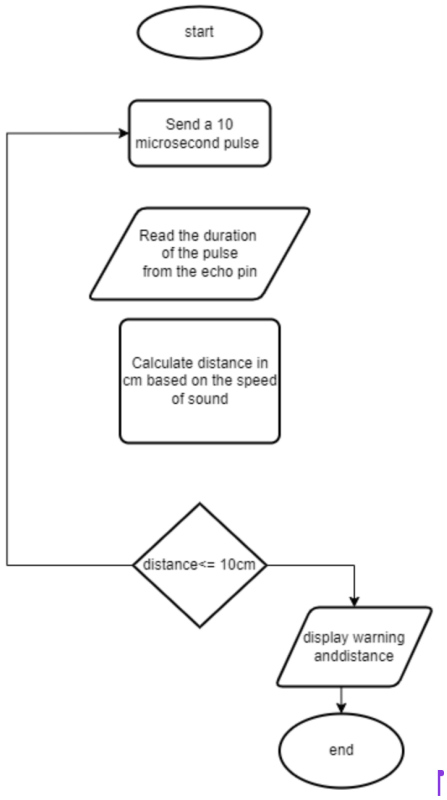


Fig. 6. Obstacle detection flowchart using ultrasonic sensor.

when the lid is closed. This mechanism is used to identify if the fuel tank is being accessed without permission, and to help prevent fuel theft.

As shown in Figure 6, the obstacle detection mechanism with ultrasonic sensor is shown. The sensor emits ultrasonic waves and calculates the time it takes to receive the reflected waves [6]. This time interval is used by the system to determine the distance to close objects. If any obstacle is identified within the set distance of 10 cm, the system will show an alarm message and the distance measured on the LCD screen.

III. Implemented Hardware System

A. List of the Hardware and Software Environment

Equipment Arrangement:

- **NodeMCU Mega WiFi R3:** Main computing unit interfacing the ATmega2560 core and the ESP8266 Wi-Fi transceivers.
- **MPU-6050 IMU:** 6-axis gyroscope and accelerometer used for sudden deceleration and collision sensing.
- **Ultrasonic Sensor (HC-SR04):** Positioned for rear obstacle detection and perimeter proximity measurements.
- **Micro Limit Switch:** Mechanically positioned to monitor the fuel tank lid aperture.
- **Hall Effect Sensor Switch:** Magnetic pulse sensor used to compute instantaneous wheel rotation speed.
- **Motor Driver (L298N):** Dual H-bridge driver delivering current to four DC gear drive motors.

Program Execution:

- **Blynk App Integration:** Configured with virtual pin triggers to direct movement and stream real-time telemetry (fuel flap state, notifications).
- **Sensor Processing Pipeline:** Reads multi-axis inertial data, pulse timing counters, proximity reflection intervals, and discrete switch logic.
- **LCD Screen Display:** Direct 16×2 parallel/I2C character display showing speed, obstacles, and diagnostic alerts locally.
- **Motor Actuation Logic:** Directional dispatch interpreting Wi-Fi telemetry into pulse-width modulated (PWM) motor driver controls.

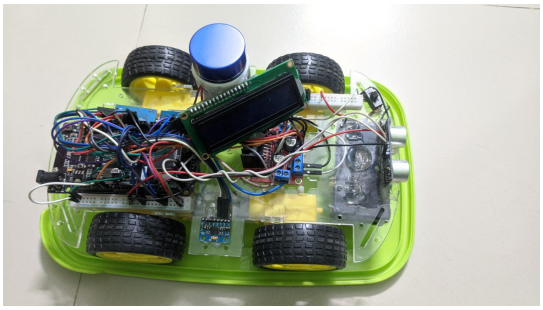


Fig. 7. Complete physical prototype of the Car Security and Monitoring System.



Fig. 8. System LCD status display while moving with fuel lid closed.

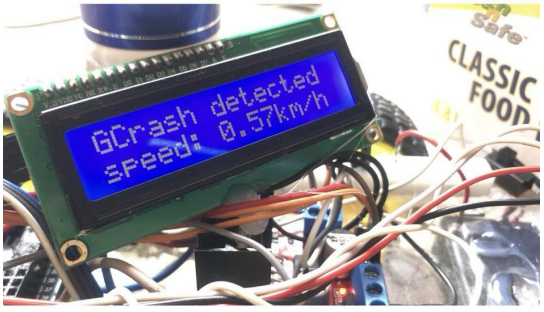


Fig. 9. LCD alert indicating collision detected by the MPU-6050 sensor.

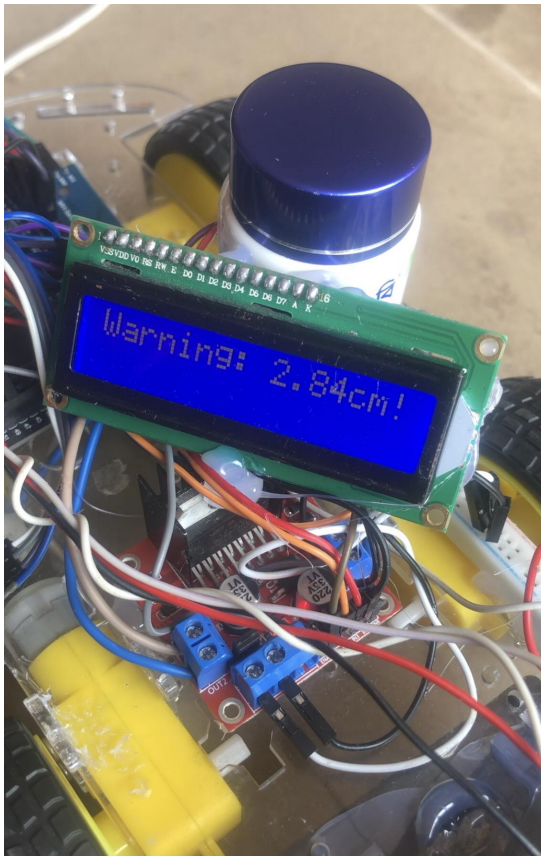


Fig. 10. LCD obstacle proximity alert displaying measured distance.

IV. Result

The Smart Car Security and Monitoring System successfully integrated all primary sensor subsystems and controller interfaces [7]. Key operational results include:

- **Fuel Theft Control:** The micro limit switch reliably detects changes in the fuel lid position and triggers alerts in the Blynk mobile interface.
- **Collision Detection:** The MPU-6050 sensor detects abrupt axis shifts and high acceleration differentials, triggering visual alert flags on the LCD screen (Fig. 9).
- **Real-Time Speed Tracking:** The Hall Effect magnetic sensor accurately computes rotational frequency and relays vehicle velocity to the display and IoT app.
- **Remote Access and Control:** Low-latency Wi-Fi telemetry enables real-time remote commands and status monitoring via the Blynk app.
- **Obstacle Detection:** The ultrasonic module flags targets within a 10 cm range, updating distance readouts in real time.

Performance Evaluation: Further quantitative testing under variable conditions is recommended to assess response latencies and sensor threshold sensitivity for severe crash events and anti-theft triggers.



Fig. 11. LCD status indicating fuel lid open condition.

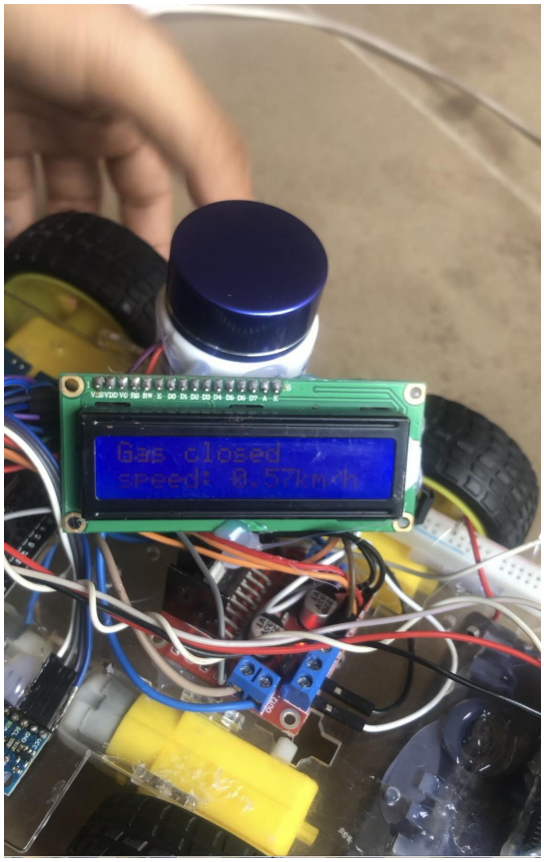


Fig. 12. LCD status indicating fuel lid closed condition.

TABLE I
COMPARISON BETWEEN PROPOSED SCHEMATIC AND PHYSICAL IMPLEMENTATION

Subsystem	Proposed Schematic	Final Hardware Implementation
Obstacle Detection	Omitted from schematic diagram	HC-SR04 ultrasonic sensor integrated for rear sensing
Emergency Alert (SOS)	SIM800L GSM module connected	Omitted due to network limitations; migrated to Blynk Wi-Fi alerts
Speed Sensing	Slotted optical IR sensor module	Hall Effect magnetic sensor module implemented

V. DISCUSSION

The working prototype demonstrates core safety functions: crash detection, wheel speed telemetry, anti-siphoning fuel lid alerts, and close-proximity obstacle warnings.

A. Limitation

Initially, the prototype included a SIM800L GSM cellular module that would be able to send cellular SMS and phone SOS messages in case of an accident. In the test environment, however, the module had regular issues connecting to the network and registering. Thus, the ability to send notifications to the emergency service was limited to the Blynk IoT interface with Wi-Fi.

B. Future Work

Future revisions of this platform will focus on:

- GPS geofencing technology to notify owners when the vehicle goes out of a specific area.
- Real-time deep learning driver profile analytics, to monitor aggressive braking and erratic driving.
- Installation of a dependable 4G-LTE based telematics transponder for the comms layer of the system. [8]

P1	p2	p3	p4	p5	p6	p7
☑	☑	☑				

Fig. 13. Complex engineering problem mapping.

VI. CONCLUSION

This paper is introducing an integrated Car Security and Monitoring System that tackle the issue of vehicle security and fuel theft. The physical prototype combines collision sensing, wheel speed calculation, fuel lid monitoring and obstacle avoidance. The experimental results confirm the potential of low-cost IoT devices to enhance the safety of passengers and mitigate car theft.

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